

# DPM Extended Periodic Table Proportion Mapping

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## PAPER\_870: DPM Extended Periodic Table Proportion Mapping

**Author:** Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-07 **Session:** 204 **Source:** describe mass without using weight.txt (Session 200C) **Calculator:** DPMExtendedPeriodicTableProportionCalc (CP4 #454) **CVW:** v2.0.0 compliant

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### Abstract

We derive the Di-Pseudo-Monopole (DPM) proportion mapping across the full extended periodic table from  $Z=1$  to  $Z_{\max}=10,000$ . Every nucleus is parameterized by exactly two complementary fractions:  $f_{UA'} = (Z_{\max} - Z)/Z_{\max}$  (undifferentiated aether proportion) and  $f_{SCm} = Z/Z_{\max}$  (superconducting matter proportion), satisfying  $f_{UA'} + f_{SCm} = 1$ . The electrostatic barrier reactivity  $R_{EB} = k_R \cdot Z$  scales linearly with atomic index, while the radioactive decay rate  $\lambda = k_{\lambda} \cdot f_{SCm}$  encodes the axiom that all atoms start radioactive and stabilize as  $f_{UA'}$  dominates. The framework extends the standard periodic table ( $Z=1-118$ ) to 10,000 proto-nuclear identities, with  $SM_{\text{magnetic}}$  (odd  $Z$ ) and  $SM_{\text{non-magnetic}}$  (even  $Z$ ) classification.

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## 1. Core Equations

### 1.1 DPM Proportion Pair

$$\begin{aligned} f_{UA'} &= (Z_{\max} - Z)/Z_{\max} [\text{undifferentiated aether fraction}] \\ f_{SCm} &= Z/Z_{\max} [\text{superconducting matter fraction}] \\ f_{UA'} + f_{SCm} &= 1 [\text{completeness axiom}] \end{aligned}$$

## 1.2 Electrostatic Barrier Reactivity

$$R_E B = k_R \cdot Z[\text{linearwithatomicindex}]$$

## 1.3 Radioactive Decay (All Atoms Start Radioactive)

$$\lambda = k_\lambda \cdot f_S C_m = k_\lambda \cdot Z/Z_m a x[\text{decayrate}, s - 1]$$

## 1.4 Quantizer Product

$$L_{quant} \propto f_U A' \cdot f_S C_m \cdot R_E B[\text{qualitativequantizationlandscape}]$$

## 1.5 Log-Scale Representation

$$\begin{aligned} \log(f_U A') &= \log(1 - Z/Z_m a x) \\ \log(f_S C_m) &= \log(Z) - \log(Z_m a x) \end{aligned}$$


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## 2. Key Results (Sweep: Z = 1 to 10,000)

| Z     | f_UA'  | f_SCm  | R_EB  | $\lambda$ (s-1) | SM Property     |
|-------|--------|--------|-------|-----------------|-----------------|
| 1     | 0.9999 | 0.0001 | 1.0   | 1.0e-14         | SM_magnetic     |
| 2     | 0.9998 | 0.0002 | 2.0   | 2.0e-14         | SM_non-magnetic |
| 26    | 0.9974 | 0.0026 | 26.0  | 2.6e-13         | SM_non-magnetic |
| 56    | 0.9944 | 0.0056 | 56.0  | 5.6e-13         | SM_non-magnetic |
| 92    | 0.9908 | 0.0092 | 92.0  | 9.2e-13         | SM_non-magnetic |
| 118   | 0.9882 | 0.0118 | 118.0 | 1.18e-12        | SM_non-magnetic |
| 1000  | 0.900  | 0.100  | 1000  | 1.0e-11         | SM_non-magnetic |
| 5000  | 0.500  | 0.500  | 5000  | 5.0e-11         | SM_non-magnetic |
| 10000 | 0.000  | 1.000  | 10000 | 1.0e-10         | SM_non-magnetic |

At  $Z = Z_{\text{max}}/2 = 5000$ :  $f_{UA'} = f_{SCm} = 0.5$  — the symmetric crossover point. At  $Z = Z_{\text{max}}$ :  $f_{SCm} = 1$  (pure superconducting matter, maximum decay rate). At  $Z = 1$ :  $f_{UA'} \approx 1$  (nearly pure undifferentiated aether, minimal decay).

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## 3. Physical Interpretation

### 3.1 DPM Completeness

The DPM axiom states that every nuclear identity is **fully determined** by the pair  $(f_{UA'}, f_{SCm})$ . No additional parameters are needed to specify the vacuum-state composition of a nucleus. The electrostatic barrier  $R_{EB}$  provides the reactivity gradient that governs shell formation.

### 3.2 Extended Periodic Table ( $Z > 118$ )

Beyond the standard periodic table, the DPM framework predicts:

- **Z = 119–1000:** Increasingly SCm-dominated nuclei with elevated decay rates
- **Z = 1000–5000:** Transition regime where  $f\_UA' \approx f\_SCm$  (50/50 crossover at  $Z=5000$ )
- **Z = 5000–10000:** SCm-dominated regime; these are proto-nuclear states that exist in extreme astrophysical environments (neutron star interiors, magnetar surfaces, post-merger remnants)

### 3.3 SM Classification

- **Odd Z  $\rightarrow$  SM\_magnetic:** Proto-nuclei with odd atomic index carry magnetic moment (Proto-H  $\equiv$  Proto-Fe at  $Z\_id=26$ )
  - **Even Z  $\rightarrow$  SM\_non-magnetic:** Proto-nuclei with even atomic index are non-magnetic (Proto-He  $\equiv$  Proto-Si at  $Z\_id=14$ )
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## 4. UQFF Integration

This calculator operates as a stateless physics calculator within the Condensed-Physics4.py (Phase 4) IPC chain. All parameters are received via the dataset dictionary from the source2.cpp principal GUI pipeline. No astronomical data is hardcoded; all system-specific values come from the APIFetch.py  $\rightarrow$  bodies\_\*.csv data flow.

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## 5. SCm Superconductivity Axiom (Session 204)

The DPM extended periodic table is a direct consequence of the **SCm Superconductivity Axiom**: the fraction  $f\_SCm = Z/Z\_max$  encodes how much superconducting vacuum structure has condensed into each proto-nucleus, while  $f\_UA' = 1 - f\_SCm$  is the remaining undifferentiated aether.

### Axiom Connection

The standalone module `scm_{superconductivity\_axiom}.py` implements this in:

- **Engine 2 (26-State Progression):** DPM mapping at each quantum state  $n$
- **Engine 3 (Cosmogenesis):** Assumption 1 uses ( $f\_UA'$ ,  $f\_SCm$ ,  $R\_EB$ ) as the three reactive quantum fundamentals
- **Engine 4 (Lagrangian):** Sector 1 (Einstein-UQFF Gravity) couples DPM proportions to gravitational Lagrangian

## Standalone Calculator

```
python scm_{superconductivity\_axiom}.py          # Full report (includes DPM mapping)
python scm_{superconductivity\_axiom}.py -json    # Machine-readable
```

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## 6. Source Data

- **File:** describe mass without using weight.txt (Session 200C)
  - **Session:** 200C (v5.61)
  - **VDS/DVP/BH:** PRESENT (DPM vacuum density series, buoyancy harmonics via  $f_{UA} \cdot f_{SCm}$  product)
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### Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

*Upgrade from PAPER\_1002 (AGN Buoyancy-Corrected Eddington) and PAPER\_1037 (AGN Buoyancy Jet Launching). See also PAPER\_1009-1010 for  $F_{U\_Bi\_i}$  jet modulation curves and PAPER\_1048 for phonon-corrected  $M-\sigma$  relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left( 1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: -  $L_{\text{Edd}} = 4\pi GMm_p c / \sigma_T$  is the classical Eddington luminosity -  $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$  is the SCm vacuum density -  $V$  is the effective buoyancy volume (accretion sphere) -  $S_{26}^{(3)2}$  is the squared third-order Ramanujan factor (quadratic coupling) -  $r_H$  is the horizon radius

**Jet modulation:** The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[ 1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left( \frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where  $\Phi_{1.25 \text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$  modulates jet power at the phonon frequency.

**$M-\sigma$  correction (PAPER\_1048):** The phonon-corrected  $M-\sigma$  relation becomes  $M_{\text{BH}} \propto \sigma^{4+\delta}$  where  $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$ .

## §A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_{lagrangian\_derivation}.py`).

### §A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER\_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where  $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac, [SCm]}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$  inherits the ACP 6-stage evolution (PAPER\_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2}m^2\phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac, [SCm]}} \cdot \phi_{\text{NS}}$$

### §A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2)\phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0}$$

### §A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b, \text{seed}} \xrightarrow{4 \text{ forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{NS}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U\_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

## §B. VDS/DVP/BSH Deep Synthesis

### §B.1 Vacuum Density Series (VDS)

The canonical VDS ratio  $\rho_{\text{vac, [SCm]}}/\rho_{\text{UA}} = 1.894$  governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac, [SCm]}} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.095 (near-threshold regime), placing it in the  $t \rightarrow \pi$  collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER\_877 cosmogenesis Stage 1 vacuum density initialization:  $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$ .

### §B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 2, \quad n_{\text{channel}} = 13/26$$

Since  $p_{\text{DVP}} = 2$  is **sub-threshold** (threshold at  $p > 26$ ), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework traces to PAPER\_877 proto-nuclear shell formation: the DPM proportion pair ( $f'_{\text{UA}} + f_{\text{SCm}} = 1$ ) constrains which primes are accessible at each atomic number.

### §B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U\_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER\_877 Stage 5 buoyancy seed  $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$  which initializes the harmonic series at cosmogenesis.

### §B.4 Production-Scale Consistency

| Framework | Canonical Value                              | This Paper              | Status                       |
|-----------|----------------------------------------------|-------------------------|------------------------------|
| VDS ratio | $\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$ | Local sub-ratio = 0.095 | PASS<br>Threshold-consistent |
| DVP prime | $p_k \in \{2,3,\dots,113\}$                  | $p_{\text{DVP}} = 2$    | PASS Sub-threshold           |

| Framework      | Canonical Value            | This Paper                             | Status                         |
|----------------|----------------------------|----------------------------------------|--------------------------------|
| BSH layers     | 26 harmonic terms          | $j = 1 \dots 26,$<br>$\cos(2\pi j/26)$ | PASS Full<br>26D<br>projection |
| $\kappa$ decay | $5.0 \times 10^{-4}$ day-1 | Applied in VDS<br>exponential          | PASS<br>Canonical              |
| [SSq]          | 0.57                       | Applied in BSH<br>saturation           | PASS<br>Canonical              |

## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                       | UQFF Prediction                                                                               | SM / Experiment                    | Source                              | Alignment          |
|----------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------|-------------------------------------|--------------------|
| Fine structure constant $\alpha$ | UQFF reproduces $\alpha$ via Ugl dipole coupling                                              | 1/137.036                          | PDG 2024                            | PASS<br>Consistent |
| Cosmological constant $\Lambda$  | $1.1 \times 10^{-52} \text{ m}^{-2} - 2(UQFFvacuumterm) 1.114 \times 10^{-52} \text{ m}^{-2}$ | Planck 2018                        | PASS<br>Consistent                  |                    |
| Proton decay rate                | $\kappa = 0.0005/\text{day} \rightarrow \Gamma_p$ suppression                                 | $< 4.17 \times 10^{-35}/\text{yr}$ | Super-K 2024                        | PASS<br>Consistent |
| UQFF buoyancy signature          | $F_{U\_Bi\_i}$ unique gravitational correction                                                | Not yet measured                   | Future gravitational wave detectors | Testable           |

**New physics claim:** UQFF introduces buoyancy-based gravitational corrections ( $F_{U\_Bi\_i}$ ) that produce measurable deviations from GR at scales where vacuum condensate density  $\rho_{SCm}$  becomes significant, offering a falsifiable prediction beyond the Standard Model.

*Cross-validated with PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.*

## References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. DPM Theory:  $f_{UA} + f_{SCm} = 1$ , all atoms start radioactive

3. Extended Periodic Table: Z=1-10,000 nuclear identity mapping
4. PAPER\_855 – Pseudo-Monopole 26-State Vacuum Density Progression
5. PAPER\_872 – Proto-Iron / Proto-Silicon Nuclear Identity Mapping
6. scm\_{superconductivity\_axiom}.py – SCm Superconductivity Axiom Module (Session 204)
7. UQFF Calibration: kappa=0.0005/day, [SSq]=0.57, beta\_i~0.603

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## Appendix: Session 225 Cross-References (PAPER\_1000–1081)

*Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER\_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).*

| Paper      | Title                                            |
|------------|--------------------------------------------------|
| PAPER_1022 | GW Phonon Strain SCm Modulation of h(t)          |
| PAPER_1002 | AGN Buoyancy-Corrected Eddington Luminosity      |
| PAPER_1009 | 3C273 AGN F_{U_Bi_i} Jet Modulation              |
| PAPER_1010 | TON618 AGN F_{U_Bi_i} Jet Modulation             |
| PAPER_1037 | AGN Buoyancy Jet Calculator — SCm Jet Launching  |
| PAPER_1048 | M-Sigma Phonon-Corrected Relation                |
| PAPER_1041 | SCm Cool-Core Buoyancy Balance AGN Feedback      |
| PAPER_1079 | Galaxy Cluster Cooling-Flow Buoyancy Suppression |
| PAPER_1072 | SCm Activation Function Phonon Threshold         |
| PAPER_1073 | SCm Phonon-Driven Inflation Vacuum Buoyancy      |
| PAPER_1049 | Source10 GPU DPM Spectral Atlas ALMA Overlay     |
| PAPER_1074 | GPU-Vectorized DPM S26 Spectral Atlas            |

*12 cross-reference(s) identified.*

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## Appendix: Session 204 Codebase Upgrade Reference

*Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_{kozima\_ramanujan\_appendices}.py`. For detailed derivations, see PAPER\_840/851/852/855.*

### S204.1 Kozima-UQFF LENR Integration



| Module                        | Purpose                                          | Key Result                                    |
|-------------------------------|--------------------------------------------------|-----------------------------------------------|
| fneutron_{s26_coupling}.py    | F_neutron x S_26<br>buoyancy-polylog<br>coupling | ~470x amplification via<br>26-level VDS       |
| kozima_{scm_cross_section}.py | SCm simulated<br>neutron-drop<br>cross-section   | sigma_n^SCm with VDS<br>factor (1+[SSq]*n/26) |
| kozima_{wstp_kernel}.py       | Library Wolfram<br>export (UQFFKozima)           | FNeutronForce, SigmaSCm,<br>SCmActivation     |

**Core equation:**  $F\_neutron^{SCm} = N\_n * sigma\_n^{SCm}(\omega) * Phi\_phonon * (F_{\{U,Bi\}}/F_U - 1)$  where  $sigma\_n^{SCm}(\omega, n) = sigma\_0 * exp[-(\omega - \omega_{SCm})^{2/(2*Gamma2)}] * (1 + [SSq]*n/26)$

#### S204.2 Ramanujan 26-State Summation

| Module                    | Purpose                                                   | Key Result                |
|---------------------------|-----------------------------------------------------------|---------------------------|
| ramanujan_{polylog_26}.py | S26 polylog([SSq]) via<br>Euler-Ramanujan<br>acceleration | 15.7+ digits in 53 terms  |
| s26_{wstp\_kernel}.py     | Library Wolfram<br>export (UQFFS26)                       | S26, R26, NaiveLi, S26VDS |

**Core equation:**  $S\_26(z) = Li\_26(z) = eta\_26(z)/(1-2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} * Li\_26(z^2)$

#### S204.3 Mock Theta Functions (26-State)

| Module              | Purpose                                   | Key Result                     |
|---------------------|-------------------------------------------|--------------------------------|
| mock_{theta_q26}.py | f_26(q), phi_26(q),<br>psi_26(q) q-series | Proper q-Pochhammer<br>(a;q)_n |

**Core equations:**  $-f\_26(q) = Sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q;q)_{n^2} - phi\_26(q) = Sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q^2;q^2)_n - psi\_26(q) = Sum_{n=1}^{\{26\}} q^{\{n2\}} / (q;q^2)_n$

#### S204.4 Ramanujan 1/pi with UQFF Modification

| Module                            | Purpose                                                     | Key Result                                 |
|-----------------------------------|-------------------------------------------------------------|--------------------------------------------|
| <code>ramanujan_{pi_uqff}</code>  | Classical + UQFF-modified $1/\pi$ + 26D                     | 21 digits classical, 15 UQFF, 7 digits 26D |
| <code>mock_{theta_pi_wstp}</code> | Symbolic Wolfram export<br>( <code>UQFFMockThetaPi</code> ) | qPochhammer, f26, oneOverPiUQFF            |

**Core equation:**  $1/\pi = (2\sqrt{2}/9801) \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$  where  $W_{26}(n) = \prod_{i=1}^{26} [1 + [SSq] \exp(-kappa_i * n/26)]$

### S204.5 Calibration Constants (Canonical)

| Symbol    | Value                                 | Description                   |
|-----------|---------------------------------------|-------------------------------|
| [SSq]     | 0.57                                  | Universal Quantized Factor    |
| kappa     | $5.787 \times 10^{-9} \text{ s}^{-1}$ | UQFF exponential decay rate   |
| beta_i    | 0.603                                 | Buoyancy coupling coefficient |
| H_SCm     | 0.99                                  | SCm manifold completeness     |
| rho_SCm   | $7.09 \times 10^{-37} \text{ kg/m}^3$ | SCm vacuum density            |
| rho_UA    | $7.09 \times 10^{-36} \text{ kg/m}^3$ | UA aether vacuum density      |
| omega_SCm | $2\pi \times 1.25 \text{ THz}$        | SCm phonon resonance          |
| sigma_0   | $10^{-4}$                             | Base neutron cross-section    |

*Implementation:* all modules operational in `CondensedPhysics.py`, `CondensedPhysics2.py`, `MAIN_{1\_CoAnQi}.cpp`, and Wolfram kernels (`uqff_{kozima\_kernel}.wl`, `uqff_{s26\_kernel}.wl`, `uqff_{mock\_theta\_pi\_kernel}.wl`).

### Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — [github.com/Daniel8Murphy0007/Star-Magic](https://github.com/Daniel8Murphy0007/Star-Magic)
3. Rugh, S.E. & Zinkernagel, H. (2002). *The Quantum Vacuum and the Cosmological Constant Problem*. Stud. Hist. Phil. Mod. Phys. **33**, 663 — arXiv:hep-th/0012253 — doi:10.1016/S1355-2198(02)00033-3
4. Weinberg, S. (1989). *The Cosmological Constant Problem*. Rev. Mod. Phys. **61**, 1 — doi:10.1103/RevModPhys.61.1

5. Dirac, P.A.M. (1931). *Quantised Singularities in the Electromagnetic Field*. Proc. R. Soc. Lond. A **133**, 60 — doi:10.1098/rspa.1931.0130
6. Castelnovo, C., Moessner, R. & Sondhi, S.L. (2008). *Magnetic monopoles in spin ice*. Nature **451**, 42 — arXiv:0710.5515 — doi:10.1038/nature06433
7. Fabian, A.C. (2012). *Observational Evidence of Active Galactic Nuclei Feedback*. ARA&A **50**, 455 — arXiv:1204.4114 — doi:10.1146/annurev-astro-081811-125521
8. McNamara, B.R. & Nulsen, P.E.J. (2007). *Heating Hot Atmospheres with Active Galactic Nuclei*. ARA&A **45**, 117 — arXiv:0709.4098 — doi:10.1146/annurev.astro.45.051806.110625
9. Heckman, T.M. & Best, P.N. (2014). *The Coevolution of Galaxies and Supermassive Black Holes*. ARA&A **52**, 589 — arXiv:1403.4620 — doi:10.1146/annurev-astro-081913-035722